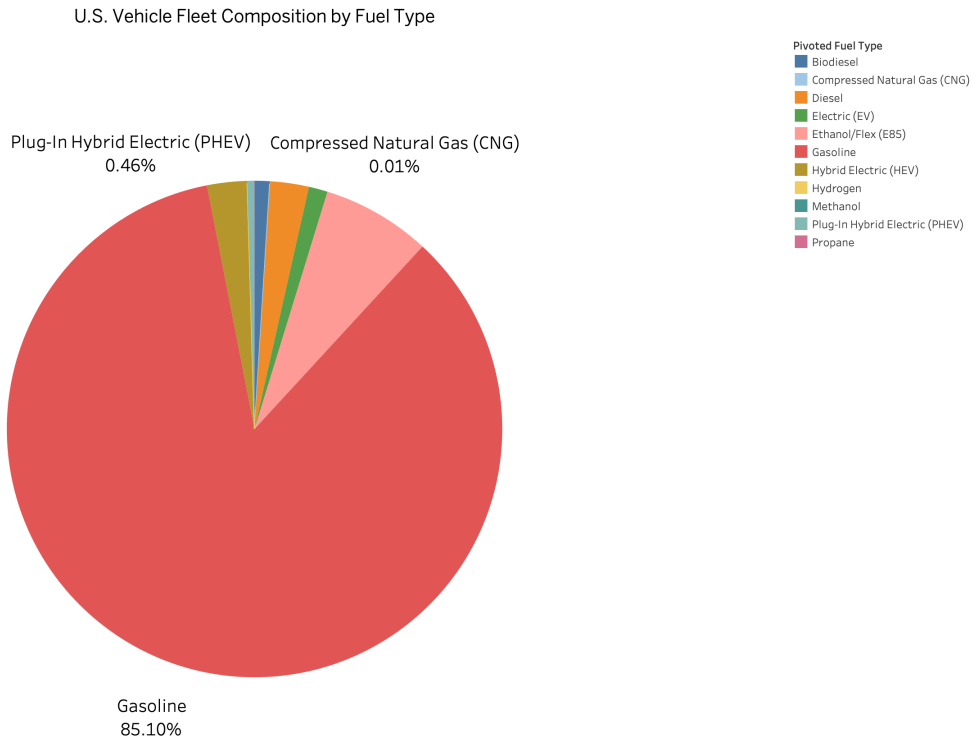
U.S. Electric Vehicle Market Share Analysis — Findings Report

1. Data Cleaning

- **Dataset**: 51 rows (50 states + District of Columbia), 13 columns
- **Fuel types**: Electric (EV), Plug-In Hybrid Electric (PHEV), Hybrid Electric (HEV), Biodiesel, Ethanol/Flex (E85), Compressed Natural Gas (CNG), Propane, Hydrogen, Methanol, Gasoline, Diesel, Unknown Fuel
- **Data quality issues addressed**: Numeric columns were stored as strings with commas (e.g., "1,256,600"). Removed commas and converted all fuel columns to float. No missing values were found. Methanol was zero across all 51 states. State names were already standardized — no duplicates or inconsistencies.
- **Total registered vehicles in the U.S.**: 287,096,500



2. Market Share Analysis

National Fuel Type Breakdown (% of all 287M vehicles)

Fuel Type	% of Total	Vehicle Count
Gasoline	84.60%	242,870,900
Ethanol/Flex (E85)	7.05%	20,240,600
Hybrid Electric (HEV)	2.58%	7,392,300
Diesel	2.50%	7,184,300
Electric (EV)	1.24%	3,555,900
Biodiesel	0.98%	2,803,600
Unknown Fuel	0.59%	1,694,100
Plug-In Hybrid (PHEV)	0.46%	1,307,200
CNG	0.009%	24,700

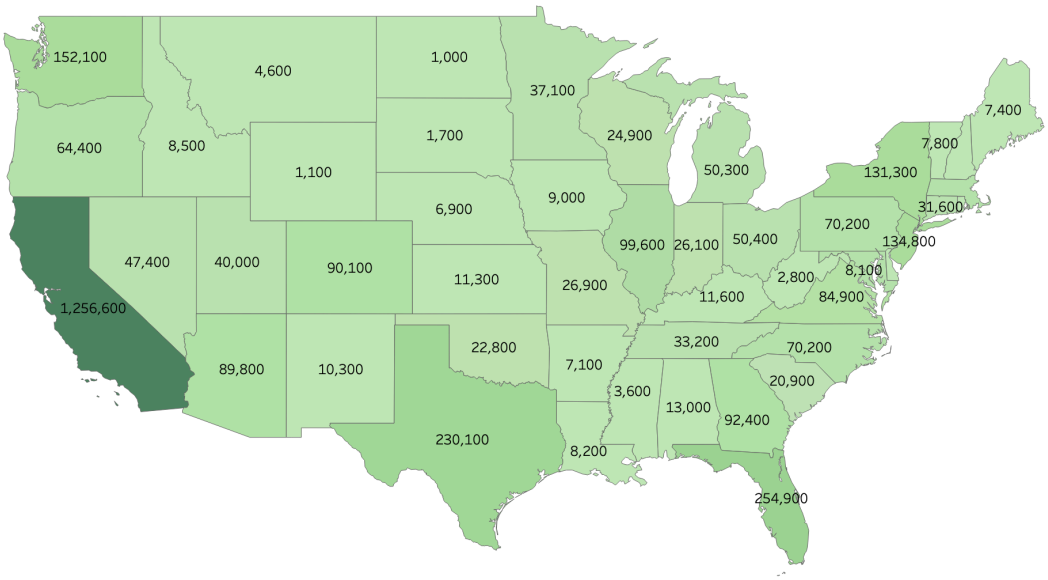
Fuel Type	% of Total	Vehicle Count
Hydrogen	0.006%	16,900
Propane	0.002%	6,000
Methanol	0.000%	0

Top 5 States by EV Adoption Rate

EVs as a percentage of each state's total registered vehicles.

Rank	State	EV %	EV Count	Total Vehicles
1	California	3.41%	1,256,600	36,850,300
2	District of Columbia	2.60%	8,100	312,100
3	Hawaii	2.37%	25,600	1,080,100
4	Washington	2.23%	152,100	6,816,100
5	Nevada	1.85%	47,400	2,556,100

Geographic distribution of EV registrations



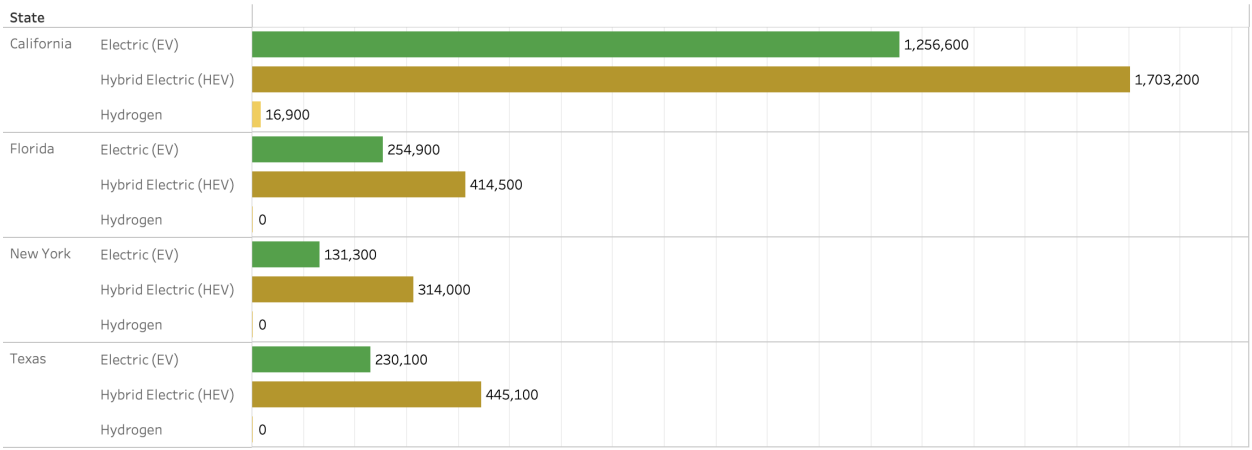
Bottom 5 States by EV Adoption Rate

Rank	State	EV %	EV Count	Total Vehicles
47	South Dakota	0.19%	1,700	918,000
48	West Virginia	0.19%	2,800	1,510,900
49	Wyoming	0.17%	1,100	652,900
50	Mississippi	0.13%	3,600	2,723,300
51	North Dakota	0.13%	1,000	797,400

California vs. Other Large States

State	EV %	PHEV %	HEV %	Gasoline %
California	3.41%	1.11%	4.62%	84.64%
Florida	1.37%	0.31%	2.23%	86.83%
New York	1.16%	0.82%	2.77%	87.52%
Texas	0.89%	0.21%	1.73%	82.93%

Alternative Electric Vehicle Registrations in California vs. Texas vs. Florida vs. New York



💡 California leads all large states by a wide margin. Its EV adoption rate (3.41%) is nearly 2.5x Florida's, 3x New York's, and nearly 4x that of Texas. California also leads in PHEV and HEV penetration, resulting in a lower gasoline dependency than New York or Florida despite having the largest total fleet.

3. Trend & Insights

Alternative Fuels: Significant vs. Niche

The three alternative fuels asked about — biodiesel, ethanol, and hydrogen — were evaluated as a percentage of total registered vehicles nationally.

Fuel	% of Total	Vehicle Count	Classification
Ethanol/Flex (E85)	7.05%	20,240,600	**Significant** — the dominant alternative fuel, larger than EVs and HEVs combined
Biodiesel	0.98%	2,803,600	**Modest/Notable** — comparable in scale to EVs (1.24%), not negligible in absolute terms
Hydrogen	0.006%	16,900	**Niche** — nearly all concentrated in California; statistically negligible nationally

For context, CNG (0.009%), Propane (0.002%), and Methanol (0%) are also niche. Combined, all three alternative fuels asked about represent 8.03% of total vehicles, but this figure is almost entirely driven by ethanol.

Key Insights

Which states are leading? California, D.C., Hawaii, Washington, and Nevada have the highest EV adoption rates. These tend to be coastal or western states with strong clean energy policies and existing charging infrastructure.

Which states lag behind? North Dakota, Mississippi, Wyoming, South Dakota, and West Virginia all have EV adoption rates below 0.2%. These states share characteristics: rural geography, lower population density, higher reliance on gasoline and diesel, and limited charging networks.

What does this imply for infrastructure planning? The gap between leading and lagging states is roughly 20x (California at 3.41% vs. North Dakota at 0.13%). This extreme disparity suggests that without targeted infrastructure investment, lagging states will fall further behind, creating a two-tier EV landscape.

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4. Recommendations: 3 States to Prioritize for EV Infrastructure Investment

1. Texas

- **EV adoption**: 0.89% (below the 1.24% national average)

- **Total fleet**: 25.8M vehicles (2nd largest in the U.S. at 8.99% of all vehicles)
- **Rationale**: Texas has the second-largest vehicle fleet in the country but an EV adoption rate well below the national average. Even a small percentage increase translates to hundreds of thousands of additional EVs. The state's high gasoline share (82.93%) and massive driving population make it a high-impact target where infrastructure investment would serve the largest possible base of potential EV adopters.

2. Florida

- **EV adoption**: 1.37% (slightly above national average, but low for its size)
- **Total fleet**: 18.6M vehicles (3rd largest, 6.47% of all vehicles)
- **Rationale**: Florida has the third-largest fleet and relatively modest EV adoption for a state of its size. Its gasoline share (86.83%) is among the highest of any large state. Florida's urban population density along the I-95 and I-4 corridors makes it well-suited for charging infrastructure buildout that could accelerate adoption significantly.

3. Georgia

- **EV adoption**: 0.96% (below national average)
- **Total fleet**: 9.6M vehicles (8th largest, 3.36% of all vehicles)
- **Rationale**: Georgia has a large fleet but below-average EV adoption, and its gasoline share sits at 85.63%. As a major logistics and transportation hub (home to the busiest airport and a major port), Georgia's infrastructure investments would have spillover effects across the Southeast, a region where EV adoption broadly lags behind coastal and western states.